Solutions to the Assignment for Week 1

Self-enforcing democracy, Ferejohn (1986), Fearon (1999)

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Remark. *These are model answers, not the only correct ones. The simulation code is in R; equivalent Python (`numpy/matplotlib`) is straightforward. For the essay questions I sketch the argument structure I would expect; depth and examples are entirely up to you.*

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1 A stylised model of self-enforcing democracy

1.1 Compliance, costs, and stakes (formal)

(a) Deriving the compliance condition and $\pi_{\min}(S)$. Recall $S \equiv R - L$ and $\kappa \equiv (1-\delta)C$. The loser complies whenever $V_E \geq V_F$:

$$\frac{L + \pi S}{1-\delta} \;\geq\; \frac{L + pS}{1-\delta} - C.$$

Multiply both sides by $1 - \delta > 0$ and cancel L:

$$\pi S \;\geq\; pS - (1-\delta)C \iff \boxed{\pi S \geq pS - \kappa}.$$

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Solving for the minimum return probability that makes the loser indifferent:

$$\pi_{\min}(S) \;=\; \frac{pS-\kappa}{S} \;=\; p-\frac{\kappa}{S}.$$

Note $\pi'_{\min}(S)=\kappa/S^2>0$: when more is at stake, the loser needs a larger institutional offer to refrain from fighting.

(b) Recurring per-period cost. If fighting incurs the cost c in every period, its present value is $\sum_{t=0}^{\infty}\delta^t c = c/(1-\delta)$, and

$$V_F^{\text{rec}} \;=\; \frac{L+pS-c}{1-\delta}.$$

Compliance requires

$$\frac{L+\pi S}{1-\delta} \geq \frac{L+pS-c}{1-\delta} \iff \pi S \geq pS-c \iff \pi_{\min}(S)=p-\frac{c}{S}.$$

The two cases differ only in scale. With a one-time cost, the per-period equivalent is $\kappa = (1-\delta)c < c$ (since $\delta>0$); recurring costs are therefore *larger* in present-value terms and pushing more strongly against fighting. Same comparative statics; bigger margin for compliance.

(c) Stake-dependent return probability $\pi(S)=\bar\pi-\gamma S$. The compliance condition becomes

$$(\bar\pi-\gamma S)\,S \;\geq\; pS-\kappa \iff -\gamma S^2+(\bar\pi-p)S+\kappa \;\geq\; 0.$$

Multiply by -1 to put it in standard form:

$$\gamma S^2-(\bar\pi-p)S-\kappa \;\leq\; 0, \qquad \text{i.e.} \qquad \gamma S^2+(p-\bar\pi)\,S-\kappa \;\leq\; 0. \quad (\star)$$

Sub-case $\gamma=0$: $(\star)$ becomes $(p-\bar\pi)S\leq\kappa$, i.e.

$$S\leq\frac{\kappa}{p-\bar\pi} \quad (\text{when } p>\bar\pi) \;\equiv\; S_{\max}.$$

This recovers the slide bound.

Sub-case $\gamma>0$: $(\star)$ is a quadratic in S with positive leading coefficient, hence concave-down upside. The roots of the equality are

$$S \;=\; \frac{-(p-\bar\pi)\pm\sqrt{(p-\bar\pi)^2+4\gamma\kappa}}{2\gamma} \;=\; \frac{(\bar\pi-p)\pm\sqrt{(\bar\pi-p)^2+4\gamma\kappa}}{2\gamma}.$$

The discriminant is strictly positive, so there are always two real roots. Since the quadratic opens upward, $(\star)$ holds between the roots. The negative root is non-positive (irrelevant for stakes), and so the binding upper bound is

$$\boxed{S^{\gamma}_{\max} \;=\; \frac{(\bar\pi-p)+\sqrt{(\bar\pi-p)^2+4\gamma\kappa}}{2\gamma}.}$$

Comparison with the slide case ($\gamma=0$). Take $\gamma\downarrow 0$ in the formula. Using $\sqrt{(\bar\pi-p)^2+4\gamma\kappa}\approx|\bar\pi-p|+\frac{2\gamma\kappa}{|\bar\pi-p|}$ when $p>\bar\pi$, the numerator behaves like $(\bar\pi-p)+(p-\bar\pi)+\frac{2\gamma\kappa}{p-\bar\pi}=\frac{2\gamma\kappa}{p-\bar\pi}$, so $S^{\gamma}_{\max}\to\kappa/(p-\bar\pi)$, recovering the linear bound. For $\gamma>0$, $S^{\gamma}_{\max}$ is strictly smaller than the linear bound: endogenously declining return probabilities tighten the self-enforcing range.

(d) Why higher stakes hurt democracy here, but help discipline in Ferejohn. Two different mechanisms:

- **Self-enforcement (this model).** The loser already accepts that the winner will collect S for one period; the question is whether her expected continuation value compensates her for waiting. As S grows, the *level* of frustration grows faster than what any fixed institutional offer $\bar{\pi}$ can compensate, especially if $\pi(S)$ itself shrinks with S. Higher S enlarges the temptation to fight more than it enlarges the value of waiting.
- **Ferejohn.** The incumbent already holds office and pays effort costs for the chance of staying in. Higher W (analogous to higher S) widens the wedge $\Delta = \delta(V^I - V^O)$, so the incumbent will pay a higher effort cost to clear the voter's bar. Higher stakes *strengthen* retrospective discipline.

The two cut in opposite directions because the actor whose incentive constraint binds is different: the *loser* in Przeworski, the *incumbent* in Ferejohn. This tension reappears repeatedly in the course (selection vs. sanctioning, sophistication vs. commitment).

1.2 Visualising self-enforcement (simulation)

```r
library(tidyverse)

## (a) primitives
delta  <- 0.90
p      <- 0.60
C      <- 1.00
pi_bar <- 0.40
kappa  <- (1 - delta) * C   # 0.10

S_grid <- seq(0.1, 2, length.out = 500)

base <- tibble(
  S        = S_grid,
  pi_min   = p - kappa / S_grid,
  pi_bar   = pi_bar
) %>%
  mutate(complies = pi_min <= pi_bar)

S_max <- kappa / (p - pi_bar)   # 0.10 / 0.20 = 0.5

## (b) plot pi_min(S) with horizontal pi_bar
ggplot(base, aes(S, pi_min)) +
  geom_line(linewidth = 1.1, colour = "steelblue") +
  geom_hline(yintercept = pi_bar, linetype = "dashed") +
  geom_ribbon(data = filter(base, complies),
              aes(ymin = pi_min, ymax = pi_bar),
              fill = "darkgreen", alpha = 0.25) +
  geom_vline(xintercept = S_max, linetype = "dotted") +
  annotate("text", x = S_max, y = 0.05,
           label = sprintf("S_max = %.2f", S_max),
           hjust = -0.05) +
  labs(x = "Stakes S", y = expression(pi[min](S)),
       title = "Self-enforcing range") +
  theme_bw()

## (c) Two facet panels: vary C and p
grid_C <- expand_grid(S = S_grid, C = c(0.5, 1, 1.5)) %>%
  mutate(kappa = (1 - delta) * C,
         pi_min = p - kappa / S)
```

```
ggplot(grid_C, aes(S, pi_min, colour = factor(C))) +
  geom_line(linewidth = 1) +
  geom_hline(yintercept = pi_bar, linetype = "dashed") +
  labs(colour = "C", title = "Vary cost of fighting C")

grid_p <- expand_grid(S = S_grid, p = c(0.4, 0.6, 0.8)) %>%
  mutate(pi_min = p - kappa / S)

ggplot(grid_p, aes(S, pi_min, colour = factor(p))) +
  geom_line(linewidth = 1) +
  geom_hline(yintercept = pi_bar, linetype = "dashed") +
  labs(colour = "p", title = "Vary force-success probability p")

## (d) Stake-dependent pi(S) = pi_bar - gamma S
gam_grid <- expand_grid(S = S_grid, gamma = c(0, 0.1, 0.2)) %>%
  mutate(pi_S   = pi_bar - gamma * S,
         pi_min = p - kappa / S)

ggplot(gam_grid, aes(S)) +
  geom_line(aes(y = pi_min), colour = "steelblue", linewidth = 1) +
  geom_line(aes(y = pi_S, colour = factor(gamma)), linewidth = 1) +
  facet_wrap(~ gamma, labeller = label_both) +
  labs(y = expression(pi), colour = "gamma",
       title = "pi_min(S) vs. pi(S) = pi_bar - gamma S")
```

Findings to report.

- Numerical $S_{\max} = 0.5$ matches the algebraic bound $\kappa/(p - \bar{\pi}) = 0.1/0.2$.
- Among C, p, varying p shifts the curve more dramatically: p enters $\pi_{\min}$ as a level shift, while C enters only through κ scaled by $(1 - \delta)$, which is small.
- For $\gamma > 0$, $\pi(S)$ and $\pi_{\min}(S)$ cross earlier as γ rises, shrinking the self-enforcing range. "High stakes" is no longer just about size: it now also means the institutional offer itself decays in S.
- *Real-world correspondences (part e).* Lowering S: constitutional limits, federalism, supermajority requirements, judicial review. Raising κ: international sanctions regimes, criminal liability for coup plotters, professional militaries with strong civilian control. Lowering p: civilian control of the military, distributed armed forces. Raising $\bar{\pi}$: open primaries, multiparty competition, peaceful track record of alternation.

1.3 Empirical illustration (essay, max. 700 words)

A high-quality answer maps the model's parameters to one concrete case. Some templates that work well:

- *US in 2020 vs. 2024 (within-country, over time).* S has risen (weakened guardrails on executive power, more aggressive use of pardons). $\bar{\pi}$ has fallen (entrenched partisan sorting reduces credible alternation in many states). κ has fallen (no successful prosecution of January 6 organisers at the apex). Net: the self-enforcing range tightened.
- *Venezuela after 2017.* Higher S (control of oil rents and judiciary), collapsing $\bar{\pi}$ (rigged elections), eroding κ (security forces aligned with incumbent). Compliance condition violated; the regime survives only by militarised repression that itself raises C for the opposition.
- *Postwar Germany 1949–present.* Designed deliberately to lower S (federalism, constructive vote of no confidence, Karlsruhe), raise κ (*wehrhafte Demokratie* bans on anti-constitutional

parties), and stabilise $\bar{\pi}$ (proportional representation with 5% threshold). Self-enforcing range is wide.

The strongest essays identify which parameter does most of the explanatory work and acknowledge what the model leaves out (legitimacy, identity, leadership).

1.4 Where does $\pi(S)$ come from? (essay, max. 1000 words)

Sample arguments by shape:

Flat (constant π): π is set by structural electoral institutions (district magnitude, threshold, vote share required). When stakes change incrementally, parties' return chances are pinned down by the rules of the game. Plausible in mature parliamentary democracies with stable electoral systems.

Increasing in S: Higher stakes mobilise the opposition's base (*negative partisanship*, donor energy, volunteer turnout), raising the opposition's effective vote share. Stylised model: opposition effort e_O is chosen to maximise expected payoff $S \cdot \pi(e_O, e_I) - c(e_O)$; as S rises, e_O^* rises and so does π. This is the standard contest-function logic.

Decreasing in S: Higher stakes raise the incumbent's incentive to invest in fraud, gerrymandering, media capture, or repression. Mechanism: incumbent manipulates the signal-to-noise ratio of elections, which directly compresses π. Examples: Hungary post-2010, Turkey post-2017, Venezuela post-2002.

Hump-shaped: At low S, "return-via-elections" is a near-irrelevance (coin-flip outcomes either way). At intermediate S, both the opposition and the incumbent have strong incentives to compete, and competitive elections naturally produce alternation around 50%. At very high S, the incumbent's incentive to manipulate eventually dominates, choking off π. This is probably the most realistic shape for partial democracies.

A good answer specifies actors, actions, payoffs, and the comparative-static inequality that delivers the proposed shape, even if the model is not fully solved. The optional formal sketch can be a Tullock contest with effort costs that depend on S asymmetrically across players.

2 Analysing the Ferejohn (1986) model

2.1 Threshold-and-shirk, with a twist (formal)

(a) The two-option structure. Given cutoff K and shock θ, only two policy choices can be optimal for the incumbent:

- Any $a < K/\theta$ delivers $u = a\theta < K$, so the incumbent is removed *and* pays positive cost. Strictly dominated by $a = 0$.
- Any $a > K/\theta$ delivers $u > K$, so the incumbent is reelected, but extra effort costs more without any return. Strictly dominated by $a = K/\theta$.

Compare the two surviving options:

$$\underbrace{W - \phi(K/\theta) + \delta V^I}_{\text{clear}} \;\geq\; \underbrace{W + \delta V^O}_{\text{shirk}} \quad\Longleftrightarrow\quad \phi(K/\theta) \;\leq\; \delta(V^I - V^O) \;\equiv\; \Delta.$$

Since ϕ is strictly increasing, $\phi(K/\theta) \leq \Delta$ is equivalent to $K/\theta \leq \phi^{-1}(\Delta)$, i.e. $\theta \geq K/\phi^{-1}(\Delta)$. Define θ^* by $\phi(K/\theta^*) = \Delta$. Clearing is optimal iff $\theta \geq \theta^*$, i.e. iff $\phi(K/\theta) \leq \Delta$. $\square$

(b) Uniform shocks, linear cost. With $F = \text{Unif}[0,1]$, $\phi(a) = a$: $\theta^* = K/\Delta$. Per-period expected performance:

$$\mathbb{E}[u] = K \cdot \Pr(\theta \geq \theta^*) + 0 \cdot \Pr(\theta < \theta^*) = K\Big(1 - \tfrac{K}{\Delta}\Big).$$

Voter maximises a quadratic in K: FOC $1 - 2K/\Delta = 0$ gives

$$K^* = \tfrac{\Delta}{2}, \qquad \theta^* = \tfrac{1}{2}, \qquad \Pr(\text{re-elect}) = \tfrac{1}{2}, \qquad \mathbb{E}[u] = \tfrac{\Delta}{4}.$$

Why tolerate shirking half the time? Tightening K extracts more on the clearing branch but raises θ^* proportionally; at the margin the gain from extra performance exactly offsets the loss from extra shirking.

(c) Quadratic cost. Now $\phi(a) = a^2/2$. The clearing condition $\phi(K/\theta^*) = \Delta$ gives $\theta^* = K/\sqrt{2\Delta}$. Per-period performance:

$$\mathbb{E}[u] = K\Big(1 - \tfrac{K}{\sqrt{2\Delta}}\Big).$$

FOC: $1 - 2K/\sqrt{2\Delta} = 0 \Rightarrow K^* = \sqrt{2\Delta}/2 = \sqrt{\Delta/2}$. Implied

$$\theta^* = \tfrac{K^*}{\sqrt{2\Delta}} = \tfrac{1}{2}, \qquad \mathbb{E}[u] = K^*(1 - \tfrac{1}{2}) = \tfrac{1}{2}\sqrt{\Delta/2} = \sqrt{\Delta/8}.$$

The shirking rate is unchanged at 1/2: this is a deeper feature of multiplicative noise plus uniform shocks. The voter's optimal compression of clearing-branch effort is symmetric in the cost shape. What does change is *how much* the voter extracts: $\mathbb{E}[u] \propto \sqrt{\Delta}$ rather than Δ. Convex costs make extracting effort harder at the margin.

(d) Why $W \uparrow$ helps Ferejohn but hurts self-enforcing democracy. W (or Δ) is the wedge between staying and losing.

- Ferejohn: a wider wedge gives the incumbent more to lose, so she will pay a larger effort cost. Voter's optimal cutoff scales with Δ, and welfare $\mathbb{E}[u]$ rises monotonically.
- Przeworski: the loser bears the wedge from the *outside*; a wider wedge makes losing more painful and waiting less appealing relative to fighting.

Same parameter, opposite signs because the affected actor's incentive constraint is different. A central tension of electoral institutions: features that strengthen sanctioning of incumbents (high salaries, long terms, broad executive powers) also raise the cost of losing for outsiders.

2.2 Simulating electoral control (simulation)

```
library(tidyverse)
set.seed(42)

simulate_ferejohn <- function(theta, K, phi, Delta) {
  # Returns a data frame of (theta, a, u, reelect)
  # Threshold-and-shirk: clear if phi(K/theta) <= Delta, else shirk.
  cost_clear  <- phi(K / theta)
  clear       <- cost_clear <= Delta
  a           <- ifelse(clear, K / theta, 0)
  u           <- a * theta            # by construction = K when clearing, 0
      else
  tibble(theta = theta, a = a, u = u, reelect = clear)
}
```

```r
T      <- 1000
Delta  <- 1
theta  <- runif(T)

## Linear cost
phi_lin <- function(a) a
K_lin   <- Delta / 2
sim_lin <- simulate_ferejohn(theta, K_lin, phi_lin, Delta)

mean(sim_lin$reelect)   # ~0.5
mean(sim_lin$u)         # ~0.25 = Delta/4

## Quadratic cost
phi_q  <- function(a) 0.5 * a^2
K_q    <- sqrt(Delta / 2)
sim_q  <- simulate_ferejohn(theta, K_q, phi_q, Delta)

mean(sim_q$reelect)     # ~0.5 again
mean(sim_q$u)           # ~ sqrt(Delta/8) ~ 0.354

## Distributions of effort and performance
bind_rows(
  mutate(sim_lin, cost = "linear"),
  mutate(sim_q,   cost = "quadratic")
) %>%
  pivot_longer(c(a, u), names_to = "var", values_to = "value") %>%
  ggplot(aes(value, fill = cost)) +
  geom_histogram(bins = 40, position = "identity", alpha = 0.5) +
  facet_wrap(~ var, scales = "free") +
  theme_bw()

## Vary Delta in {0.5, 1, 2}
results <- map_dfr(c(0.5, 1, 2), function(D) {
  K_l <- D / 2
  K_q <- sqrt(D / 2)
  bind_rows(
    simulate_ferejohn(theta, K_l, phi_lin, D) %>% mutate(cost = "linear",
        Delta = D),
    simulate_ferejohn(theta, K_q, phi_q,  D) %>% mutate(cost = "quadratic",
        Delta = D)
  )
})

results %>% group_by(Delta, cost) %>%
  summarise(reelect = mean(reelect),
            mean_a  = mean(a),
            mean_a_clear = mean(a[reelect]),
            mean_u  = mean(u),
            .groups = "drop")
```

What to take from the simulation (parts d–f). Mechanically, raising Δ raises both K and $\mathbb{E}[u]$ but leaves the shirking rate at exactly 1/2. Substantively, Δ summarises "how much it hurts to lose office" — W, term length, post-office prosecution risk all load on it. The voter's irreducible trade-off is between effort intensity per clearing period and the frequency of shirking; the wedge Δ shifts the ceiling on what is attainable.

2.3 Are stationary cutoff rules suboptimal? (essay, max. 1000 words)

Sketch of an answer engaging at least two of (a)–(d):

- *(a) Why stationarity may be too rigid.* The voter only observes performance $u = a\theta$, not the

components a and θ separately. A stationary K does not exploit information that accrues over time (e.g., the empirical distribution of past performance, the current state of the economy, public knowledge of an exogenous shock). A history-contingent K_t would let voters demand more from incumbents when conditions are easy and less when shocks are bad.

- *(b) Declining vs. increasing cutoffs.* Acharya, Lipnowski, and Ramos (2025) and Gieczewski and Li (2024) show that increasing leniency over time can sustain more effort: the incumbent's incentive constraint is hardest to satisfy in the first term (small reputation, much to prove). Backloading rewards via a weakly declining cutoff trajectory exploits this. Increasing cutoffs make sense when monitoring quality improves with tenure (e.g., longer track records).
- *(c) Incumbency advantage.* Ferejohn-style models can rationalise it: re-elected incumbents are precisely those who cleared a stringent bar, so the surviving pool is enriched in incumbents who can clear future bars cheaply (through experience, networks, name recognition). This rational-screening explanation is observationally similar to selection on quality, but distinguishable from raw partisanship by examining how incumbency advantage scales with the noisiness of θ.
- *(d) Real-world example.* Retrospective voting on the economy (the classic Fiorina/Erikson literature). Hidden action: presidential effort on macroeconomic management. Noisy signal: GDP growth, unemployment, inflation (buffeted by oil prices, currency moves, foreign demand). Stationary cutoff: voters punish below-average growth in the election year. Punishment available: removal from office.

3 Analysing the Fearon (1999) model

3.1 Selection, sanctioning, and the commitment problem (formal)

(a) Pure selection: midpoint cutoff. Good types deliver $z = -0 + \varepsilon = \varepsilon$, distributed by $f(\cdot)$. Bad types deliver $z = -x_b^2 + \varepsilon$, distributed by $f(\cdot + x_b^2)$ (i.e., shifted left by x_b^2). Posterior:

$$\Pr(\text{good} \mid z) \;=\; \frac{\alpha\, f(z)}{\alpha\, f(z) + (1-\alpha)\, f(z + x_b^2)}.$$

A re-elected challenger is, on average, "good with probability α" (the prior). So the voter reelects iff $\Pr(\text{good} \mid z) \geq \alpha$, equivalent to $f(z)/f(z + x_b^2) \geq 1$. Symmetry and unimodality of f centred at zero imply $f(z) \geq f(z + x_b^2)$ iff $|z| \leq |z + x_b^2|$. Squaring,

$$z^2 \leq (z + x_b^2)^2 \iff 0 \leq 2x_b^2 z + x_b^4 \iff z \geq -x_b^2/2.$$

Hence $k^* = -x_b^2/2$. The cutoff lies exactly halfway between the good type's mean signal (0) and the bad type's mean signal ($-x_b^2$): symmetry makes equal-likelihood-ratio occur at the midpoint of the two means.

(b) Pure sanctioning: deriving $k^* = -x^2$. With $z = -x^2 + \varepsilon$, $\Pr(z \geq k \mid x) = \Pr(\varepsilon \geq k + x^2) = 1 - F(k + x^2)$. The bad incumbent solves

$$\max_x \; W - (1-x)^2 + \delta W\big[1 - F(x^2 + k)\big].$$

FOC:

$$2(1-x) - \delta W \cdot 2x \cdot f(x^2 + k) = 0 \iff 1 - x = \delta W\, x\, f(x^2 + k). \quad (\dagger)$$

The voter wants to maximise the marginal sensitivity of reelection probability to x at the equilibrium. That marginal sensitivity is $-2xf(x^2 + k)$, maximised in absolute value when f is at its mode. By unimodality f peaks at zero, so the voter sets k such that $x^2 + k = 0$, i.e. $k^* = -x^2$. The density at the equilibrium is therefore $f(0)$, "the most informative point".

(c) Solving for x^* and the role of $f(0)$. Substitute $k^* = -x^2$ into (†), so $f(x^2 + k^*) = f(0)$:

$$1 - x = \delta W \cdot x \cdot f(0) \iff x^* = \frac{1}{1 + \delta W f(0)}, \qquad k^* = -(x^*)^2.$$

Comparative statics in $f(0)$. A higher density at the mode (less noisy information) makes the reelection probability more sensitive to x. The bad incumbent moderates more (smaller x^*). Intuition: better monitoring sharpens the punishment, so the incumbent has more reason to pander. Same direction for δ and W.

(d) The mixed case and the commitment problem. With $\alpha > 0$, the voter's selection logic (from part a) pulls the cutoff to $k = -x_b^2/2$ rather than $-x_b^2$. Plug this into the bad type's FOC:

$$1 - x_b = \delta W \cdot x_b \cdot f(x_b^2 + k) = \delta W \cdot x_b \cdot f(x_b^2/2).$$

Since f is unimodal at zero, $f(x_b^2/2) < f(0)$ for any $x_b \neq 0$. The bad type's marginal incentive to moderate is strictly weaker than under pure sanctioning. Solving, $x_b^M > x^S$.

Commitment problem in words. Ex ante the voter would like to commit to $k = -x_b^2$ (placing f at its mode) so the bad incumbent shirks the least possible. Ex post, after seeing the signal, Bayes' rule tells the voter something useful about *type*, and rationally she wants to use it — which means setting $k = -x_b^2/2$, the selection-optimal cutoff. Because the voter cannot bind her future self to ignore type information, the sanctioning-optimal rule is not implementable. Even a sliver of good types ($\alpha > 0$) is enough to trigger this collapse, which is why the equilibrium is discontinuous at $\alpha = 0$.

3.2 Simulating the selection–sanctioning trade-off (simulation)

```
library(tidyverse)
set.seed(7)

W     <- 1
delta <- 0.9
sigma <- 0.5
f0    <- dnorm(0, 0, sigma)        # density at zero

## (a) pure-sanctioning benchmark
xS <- 1 / (1 + delta * W * f0)
kS <- -xS^2
xS; kS

## (b) pure selection with x_b = 0.5
xb <- 0.5
kP <- -xb^2 / 2

n  <- 1000
good <- rnorm(n, mean = 0,    sd = sigma)
bad  <- rnorm(n, mean = -xb^2, sd = sigma)
mean(good >= kP)              # good-type reelection rate
mean(bad  >= kP)              # bad-type reelection rate
share_bad <- sum(bad >= kP) / sum(c(good, bad) >= kP)
share_bad                      # share of reelected who are bad

## (c) mixed-case approximate fixed point
xgrid <- seq(0, 1, length.out = 1001)
br_curve <- function(x_hat) {
  k <- -x_hat^2 / 2
  payoff <- W - (1 - xgrid)^2 +
            delta * W * (1 - pnorm((k + xgrid^2) / sigma))
  xgrid[which.max(payoff)]
```

```
}
br <- map_dbl(xgrid, br_curve)

## fixed point: where br(x_hat) is closest to x_hat
xM <- xgrid[which.min(abs(br - xgrid))]
xM

## (d) simulate with the mixed-case x_b = xM
kM   <- -xM^2 / 2
goodM <- rnorm(n, mean = 0,     sd = sigma)
badM  <- rnorm(n, mean = -xM^2, sd = sigma)
mean(goodM >= kM); mean(badM >= kM)

## (e) repeat for sigma in {0.25, 0.5, 1}
out <- map_dfr(c(0.25, 0.5, 1.0), function(s) {
  f0_s <- dnorm(0, 0, s)
  xS_s <- 1 / (1 + delta * W * f0_s)
  br_s <- function(x_hat) {
    k <- -x_hat^2 / 2
    payoff <- W - (1 - xgrid)^2 +
              delta * W * (1 - pnorm((k + xgrid^2) / s))
    xgrid[which.max(payoff)]
  }
  br_vals <- map_dbl(xgrid, br_s)
  xM_s    <- xgrid[which.min(abs(br_vals - xgrid))]
  tibble(sigma = s, xS = xS_s, xM = xM_s, gap = xM_s - xS_s)
})
out
```

Findings to report (part f). Across noise levels σ, $x^M > x^S$ holds robustly, and the gap widens as σ falls (cleaner signals make selection more attractive, but the sanctioning benchmark also tightens; the difference is the strength of the commitment problem). The incentive to moderate weakens *at the equilibrium density* $f(x_b^2/2)$: that is the "where exactly" Fearon's argument points to.

3.3 Noise structures and voter mistakes (formal + simulation + substantive)

(a) Why the midpoint cutoff? Good type plays $x_g = 0$, so $z_g = \varepsilon$ has density $f(\cdot)$ centred at 0. Bad type plays x_b, so $z_b = -x_b^2 + \varepsilon$ has density $f(\cdot + x_b^2)$ centred at $-x_b^2$. With symmetric, unimodal f, the two densities are mirror images around $-x_b^2/2$. The voter's posterior odds favour "good" iff $f(z) \geq f(z + x_b^2)$, which by the calculation in 3.1(a) is $z \geq -x_b^2/2$. The midpoint is where the two density curves cross.

(b) Varying σ with normal noise.

```
library(tidyverse)
set.seed(11)
xb <- 0.5
k  <- -xb^2 / 2

n <- 1000
make_run <- function(s) {
  good <- rnorm(n, 0,     s)
  bad  <- rnorm(n, -xb^2, s)
  tibble(
    sigma         = s,
    bad_reelected = mean(bad  >= k),       # false positive
    good_removed  = mean(good <  k)        # false negative
  )
}
```

```
map_dfr(c(0.25, 0.5, 1), make_run)
```

What changes: both error rates rise with σ (more overlap of densities). What does not change: the cutoff stays at the algebraic midpoint, and by symmetry the two error rates remain *equal* for any σ when x_b is held fixed.

(c) Skewed noise. With shifted log-normal ε (mean zero, right-skewed), good and bad densities are no longer mirror images around $-x_b^2/2$. The midpoint cutoff no longer equates the two errors: the long upper tail of good types pushes their density above the bad density at the midpoint, but the same skew also makes the bad density's right tail reach further. Net: false positives and false negatives become asymmetric. A welfare-maximising cutoff would shift to the heavy-tail side. Skewness matters because the cutoff's optimality relied on the symmetry of f – without it, midpoint $\neq$ likelihood-ratio crossing.

(d) Multiplicative noise. With $z = -x^2\varepsilon$ and lognormal $\varepsilon > 0$:

- Good types ($x = 0$) deliver $z = 0$ deterministically.
- Bad types deliver $z = -x_b^2\varepsilon < 0$, with magnitude scaled by ε.

The bad-type signal distribution is no longer a translation of the good-type distribution; it is a scaling. A single cutoff in z separates types perfectly when good types deliver $z = 0$ exactly, but in any realistic extension (additive noise on top), the variance of bad-type signals depends on x_b, and the likelihood ratio is no longer monotone in z. A single cutoff may misclassify in both tails.

(e) Substantive interpretation.

- *Symmetric additive noise*: well-functioning routine policy with small idiosyncratic shocks (e.g., monthly unemployment). Voters can use a midpoint rule.
- *Skewed shocks*: policies with rare-but-large downside (financial crises, natural disasters) or rare-but-large upside (windfalls). Voters who ignore skewness will systematically over- or under-react.
- *Multiplicative noise*: policy domains where competence and luck interact (national-security, foreign policy: a competent leader makes good luck better and bad luck less bad). Single cutoffs miss the variance signal; voters should use richer rules (or trust expert intermediaries).

3.4 Ignorance, sophistication, and voter commitment

(a) Restating Fearon's commitment problem. Voters would like to commit ex ante to $k^P = -x_b^2$ (ignore type, place f at its mode). Ex post, after observing z, they cannot resist the temptation to use z to update about type, which delivers $k^S = -x_b^2/2$. The sanctioning-optimal cutoff is not implementable.

(b) Ignorance as commitment. An ignorant voter does not update on z for the purpose of inferring type. Mechanically, she still applies a cutoff (otherwise she could not vote), but that cutoff is detached from the type-inference logic. The ignorant cutoff k^I serves only sanctioning purposes. So ignorance *commits* the voter to the ex-ante optimal rule — not because she is wiser, but because she cannot do the inference Bayes would prescribe.

(c) The two effects of sophistication.

1. *Information effect*: sophisticated voters distinguish noise from type, leading to better selection of good types when α is intermediate.
2. *Commitment effect*: sophisticated voters cannot help but use type information, undermining the sanctioning incentive.

The two cut against each other for accountability; whether sophistication helps overall depends on the relative weight on selection vs. sanctioning.

(d) Majority rule with two voter types. Let β be the share of sophisticated voters. Given $k^I < k^S$:

- $z < k^I$: both types remove. Incumbent loses regardless of β.
- $k^I \leq z < k^S$: ignorant retain, sophisticated remove. Outcome by majority: incumbent wins iff $\beta < 1/2$.
- $z \geq k^S$: both types retain. Incumbent wins regardless of β.

(e) Effective cutoff. Reading off (d):

$$k^{\text{eff}}(\beta) = \begin{cases} k^I, & \beta < 1/2, \\ k^S, & \beta > 1/2. \end{cases}$$

The bad type's incentive to choose low x_b *drops discretely* as the electorate crosses from ignorant majority to sophisticated majority: the reelection cutoff moves from sanctioning-optimal to selection-optimal, weakening the discipline that the bad type faces. Discontinuity at $\beta = 1/2$.

(f)/(g) Simulation outline.

```r
library(tidyverse)
set.seed(13)

W     <- 1
delta <- 0.9
sigma_const <- 0.5

xgrid  <- seq(0, 1, length.out = 1001)
betas  <- c(0, 0.25, 0.49, 0.51, 0.75, 1)

best_response <- function(x_hat, beta, sigma) {
  kI <- -x_hat^2
  kS <- -x_hat^2 / 2
  k_eff <- if (beta < 0.5) kI else kS
  payoff <- W - (1 - xgrid)^2 +
            delta * W * (1 - pnorm((k_eff + xgrid^2) / sigma))
  xgrid[which.max(payoff)]
}

fixed_point <- function(beta, sigma) {
  br <- map_dbl(xgrid, ~ best_response(.x, beta, sigma))
  xgrid[which.min(abs(br - xgrid))]
}

## (f) constant sigma
res_f <- tibble(
  beta   = betas,
  xb_star = map_dbl(betas, ~ fixed_point(.x, sigma_const))
```

```
)

## (g) sigma(beta) = 1 - 0.5 * beta -- sophistication improves signal quality
res_g <- tibble(
  beta  = betas,
  sigma = 1 - 0.5 * betas,
  xb_star = map2_dbl(betas, sigma, fixed_point)
)

bind_rows(
  mutate(res_f, scenario = "constant sigma"),
  mutate(res_g, scenario = "sigma(beta) = 1 - 0.5 beta")
) %>%
  ggplot(aes(beta, xb_star, colour = scenario)) +
  geom_point(size = 3) + geom_line() +
  geom_vline(xintercept = 0.5, linetype = "dashed") +
  labs(y = expression(x[b]^"*"), title = "Mixing ignorant and sophisticated
      voters")
```

(h) Putting the pieces together.

- Sophistication helps when α (good-type prior) is intermediate and σ falls fast enough with β that the informational gain dominates. Selection of good types is the main channel.
- Sophistication hurts when bad types dominate the pool or when σ is relatively insensitive to β. The commitment cost dominates: the bad type shirks more, knowing the cutoff is selection-optimal.
- Net: sophistication is good for accountability when its informational benefits scale with it (as in (g)), bad otherwise.

(i) Pushing back on the binary. Real voters lie on a spectrum, and parties supply cues that compress this heterogeneity. Three concrete amendments:

- *Partial sophistication*: voters update partially, e.g., a shrinkage estimator. Cutoff lies between k^I and k^S, smoothing the discontinuity.
- *Party cues*: parties commit to retention rules (e.g., primaries that remove low-performing incumbents), supplying the commitment voters lack.
- *Institutional pre-commitment*: term limits, mandatory audits, fixed KPIs — all mechanically commit the electorate to a sanctioning rule that ignores type. Cost: foregone selection benefits.

3.5 Selection versus sanctioning (essay, max. 900 words)

Sample structure:

(a) Selection vs. sanctioning. Selection: voters choose *who* governs, given fixed type. Voters observe a signal of type and decide whether to retain. Sanctioning: voters discipline *actions*, given identical politicians. They observe performance and decide whether to remove. Difference: selection works on the type pool; sanctioning works on the incentive constraint of the marginal incumbent.

(b) Why selection fits real-world behaviour better. Empirical patterns that pure sanctioning cannot easily explain: dislike of office-seekers, support for term limits, premium on consistency, absence of last-period shirking.

(c) The commitment problem in plain words. If voters can spot good types from the data, they will use that information at the polls — even when they would, ex ante, have benefited from *not* using it. Bad types know this and respond by shirking more, since the punishment they face is no longer sanctioning-optimal.

(d) Selection vs. sanctioning by domain. *Corruption*: selection works better in low-information environments where the voter cannot distinguish corruption from incompetence; sanctioning bites when scandals are formal and well-publicised. *Ideological extremism*: selection wins (Hall 2015 RDD on US primaries). *Economic management*: sanctioning works in attribution-clean environments (small open economies with visible exchange-rate movements); selection helps in noisy ones (large diversified economies subject to many shocks).

3.6 Are term limits good for accountability? (essay, max. 1000 words)

Sketch covering all five points:

(a) Reelection creates the wedge Δ (Ferejohn) and the future-period discount δW (Fearon). Both are the prizes for compliance. In the final period, the wedge collapses to zero — no future, no incentive to clear any cutoff. Pure shirking.

(b) With noisy signals, a few terms may be too short for voters to confidently identify and remove a bad type. Term limits force replacement after a fixed horizon, providing a mechanical sanction even when the informational environment is too poor to support inference-based removal. The cost: good types are forced out as well.

(c) Costs: removed effort incentive in last term (Ferejohn), loss of expertise and policy continuity (lame ducks), risk of ousting good types, strategic positioning by would-be successors. Long-term projects that need continuity (constitutional reform, infrastructure) suffer most.

(d) Term limits are most beneficial when: (i) information is so noisy that voters cannot distinguish types, (ii) entrenchment risk is high (clientelism, incumbent capture of media), (iii) institutions can absorb the cost of forced turnover. Most damaging when: (i) the labour pool of competent politicians is shallow, (ii) good types are rare, (iii) the marginal incumbent is highly responsive to electoral incentives (high-Δ sanctioning regimes).

(e) Better information substitutes for term limits. Robust audits, independent media, and credible policy-performance metrics let voters distinguish types directly, shrinking the case for mechanical replacement. The case for term limits is weakest precisely where direct accountability is strongest. (Note also that term limits and the loss-of-information story interact: better information makes the foregone selection benefit of term limits larger.)

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